REPORT WRITING

Course: Application of Information & Communication Technologies
Program: BS Information Technology (1st Semester)
Topic: Software and Computer Security Risk
Total Marks: 10
Weightage: 10% of Final Assessment

INSTRUCTIONS FOR STUDENTS

1. This is an **individual assignment**. Plagiarism or copying from other students will result in a zero mark.
2. The report should be **typed** (MS Word) using the following format:
    - Font: Times New Roman / Arial
    - Font Size: 12pt for body text, 14pt for headings
    - Line Spacing: 1.5
    - Page Limit: **Minimum 5 pages, Maximum 8 pages** (excluding title page and references)
3. Submit the report in both **printed form** (to your instructor) and **soft copy** (to CR/GR).
4. Use at least **3 credible references** (books, research papers, or reputable websites). Wikipedia is not acceptable.
5. Include a **references list**.

REPORT TOPIC: "The Dual Nature of Software: Productivity Tool vs. Security Threat"

REPORT OBJECTIVES

This report aims to:

- Develop a deep understanding of software types and their functions.
- Analyze the relationship between software behavior and computer security risks.
- Evaluate real-world malware incidents and their impact.
- Propose practical security measures for individuals and organizations.
- Connect theoretical concepts from the lectures to real-world scenarios.

REPORT STRUCTURE AND GUIDELINES

Your report must include the following sections in the order given:

SECTION 1: TITLE PAGE

Include:

- Report Title
- Your Name
- Your Roll Number

- Course Name and Code
- Instructor's Name
- Submission Date

SECTION 2: INTRODUCTION

Write a brief introduction (150-200 words) that:

- Defines software and its importance in modern computing.
- Introduces the concept of computer security risks.
- States the purpose and scope of your report.
- Provides a roadmap of what the reader will find in the report.

SECTION 3: UNDERSTANDING SOFTWARE

In this section, you must demonstrate your understanding of software concepts.

Address the following points:

1. Define **System Software** and **Application Software**. Provide at least **three examples** of each.
2. Explain the role of **Utility Programs** in system maintenance. Discuss at least **three utility programs** mentioned in the lecture (e.g., File Manager, Disk Defragmentation, Backup Utility, File Compression).
3. Differentiate between **Shareware, Freeware, and Licensed Software**. Why is understanding software licensing important for ethical computing?

SECTION 4: COMPUTER SECURITY RISKS – AN OVERVIEW

In this section, you must explain the various security risks.

Address the following points:

1. Define **computer security risk** and **cybercrime**.
2. List and briefly explain the **six types of computer security risks** shown in the diagram on page 3 of ICT 08. (Internet/network attack, unauthorized access, hardware theft, software theft, information theft, system failure)
3. What are the common **symptoms of an infected computer**? List at least **eight symptoms** from the lecture.

SECTION 5: TYPES OF MALICIOUS SOFTWARE – DETAILED ANALYSIS

This is the core section of your report. You must provide a detailed analysis of different types of malware.

For each of the following malware types, provide:

- A clear definition.
- How it works (mechanism of infection/attack).
- What damage it can cause.
- A real-world example (from history or recent news).

Malware types to cover:

1. **Virus** (including Boot Sector Virus, Chernobyl Virus, Logic Bomb/Time Bomb, Memory Resident Virus)
2. **Worm**

3. **Trojan Horse**
4. **Rootkit**
5. **Ransomware**
6. **Spyware / Adware**

SECTION 6: CASE STUDY ANALYSIS – A REAL-WORLD MALWARE ATTACK

Choose **ONE** real-world malware attack from the list below and write a detailed case study:

- **ILOVEYOU Worm (2000)**
- **Stuxnet (2010)**
- **WannaCry Ransomware (2017)**
- **Petya/NotPetya (2017)**

Your case study must include:

- Background: When and where did it occur?
- Type of Malware: Which category from Section 5 does it belong to?
- Infection Vector: How did it spread?
- Damage Caused: What was the impact (financial, operational, social)?
- Lessons Learned: What security measures could have prevented it?

SECTION 7: PROTECTION AND PREVENTION STRATEGIES

Address the following:

1. The role of **Antivirus Software**: How does it work? Why is it important to keep it updated?
2. The importance of **Firewalls**: What do they protect against?
3. **Backup**: Why is it called "The Ultimate Safeguard"? Explain full vs. selective backup.
4. **Safe Practices**: List at least **five safe computing practices** for everyday users (e.g., not opening suspicious emails, scanning removable media, avoiding pirated software).

SECTION 8: CONCLUSION

Summarize the key points of your report in 150-200 words. Reflect on the importance of understanding both software and security risks for an IT professional in Pakistan.

SECTION 9: REFERENCES

List all sources you used in preparing this report. Use a consistent citation style (APA, MLA, or IEEE). At least **three credible sources** are required.

Example:

- Vermaat, M. E. (2018). *Discovering Computers & Microsoft Office 365*. Cengage Learning.
- Symantec Corporation. (2023). *Internet Security Threat Report*. Retrieved from www.symantec.com
- Kaspersky Lab. (2022). *What is Malware?* Retrieved from www.kaspersky.com

MARKING RUBRIC

Section	Description	Marks
Section 2	Introduction – clarity, scope, roadmap	1
Section 3	Understanding Software – accuracy, examples, explanations	2
Section 4	Security Risks Overview – completeness, accuracy	2
Section 5	Malware Analysis – depth, clarity, real-world examples	2
Section 6	Case Study – relevant selection, detailed analysis, lessons learned	1
Section 7	Protection Strategies – practical, comprehensive, connected to lecture	1
Section 8 & 9	Conclusion + References – quality of summary, quality and number of sources	1
Total		**10**

ADDITIONAL GUIDELINES

- **Plagiarism Policy:** Any copied content will be detected and will result in a zero mark. Use your own words.
- **Language:** Use formal, academic English. Proofread your report before submission.

GOOD LUCK!